

1 The Pleistocene-Holocene Transition in Tagua-Tagua,
2 central Chile

3 Matías Frugone-Álvarez^{*,a}, Erwin Gonzáles Huerta^{c,1,e}, Blas Valero-Garcés^d,
4 Carolina Godoy^{c,1,e}, Josu Aranbarri^{c,1,e}, Penélope Gonzáles^f, Antonio
5 Delgado-Huerta^{c,1,e}, Claudio Latorre¹, Rafael Labarca Encina^{c,1,e}

6 ^a*Pontificia Universidad Católica de Chile, Departamento de Ecología-Centro UC Desierto de*
7 *Atacama, Av. Libertador Bernardo O'Higgins 340, Santiago, Chile.*

8 ^b*Instituto de Ecología y Biodiversidad (IEB), Santiago, Chile*

9 ^c*Instituto Pirenaico de Ecología (IPE-CSIC), Zaragoza, Spain.*

10 ^d*Centro de Investigación en Biodiversidad y Ambientes Sustentables (CIBAS), Chile*

11 ^e*Department of Geology and Environmental Science, University of Pittsburgh, Pittsburgh,*
12 *USA*

13 ^f*Instituto Andaluz de Ciencias de la Tierra (CSIC-UGR), Granada, Spain*

14 **Abstract**

Paleohydrological reconstructions from the Chilean Altiplano document abrupt moisture fluctuations during the last millennia. Although the end of the mid Holocene aridity and the onset of more humid conditions between 6–4 ka has been identified in numerous regional marine and terrestrial sites, the timing of late Holocene dry and humid phases shows large regional variability. Laguna Miscanti and Laguna Miñiques (23° 43'S – 67° 46'W, 4140 m asl) are topographically closed, but connected by surface outflow from Miscanti. Sedimentological and geochemical indicators from two new cores show large facies changes, i.e. higher carbonate and evaporite deposition during more arid periods and increased organic productivity (both algal and macrophyte) during more humid phases. As in most Andean lakes located in volcanic settings, large 14C reservoir effects occur complicating 14C dating, so the age models include 210Pb and U/Th dating. In spite of dating uncertainties, both lakes show similar patterns during the last millennium. A humid phase in Laguna Miscanti prior to ca 1200 CE is coherent with rodent middens and geomorphological features indicative of a major pluvial/recharge event at lower altitudes (Atacama Desert) during the Medieval Climate Anomaly (ca 800 - 1300 CE). The LIA (1300 – 1850 CE) is characterized by several arid/humid cycles and the last century by a productivity increase. The hydrological changes observed during the last millennium illustrate the complex dynamics of recent climate evolution over the high altitude Andean plateau. Discrepancies between the northern and southern Altiplano records and with intermediate latitudes (Central Chile) records may reflect contrasting responses to external forcing (Westerlies versus South American Monsoon dynamics) along different climatic zones.

15 *Key words:* keywords

16 *Text based on elsarticle sample manuscript, see <http://www.elsevier.com/>*

*Corresponding Author
Preprint submitted to *Quaternary Science Reviews*
Email address: matutefrugone@gmail.com (Matías Frugone-Álvarez)

July 14, 2021

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18 **1. Introduction**19 **2. Study site and regional setting**20 **3. Materials and methods**21 *3.1. Sedimentary and geochemical analysis*

22 In Dicembre 2013, we recovered 5 cores sediment at several sites around to
 23 Tagua Tagua 3 site using sing a hammer-modified UWITEC gravity corer and a
 24 UWITEC platform (Fig. 1 and Table S1). The cores were transported to the
 25 IACT-CSIC (Granada) and IPE-CSIC (Zaragoza-Jaca) laboratories in Spain for
 26 image acquisition and measurement of sediment physical properties, density and
 27 Magnetic Susceptibility (MS [$\times 10^{-8}$ (m^3/kg)])

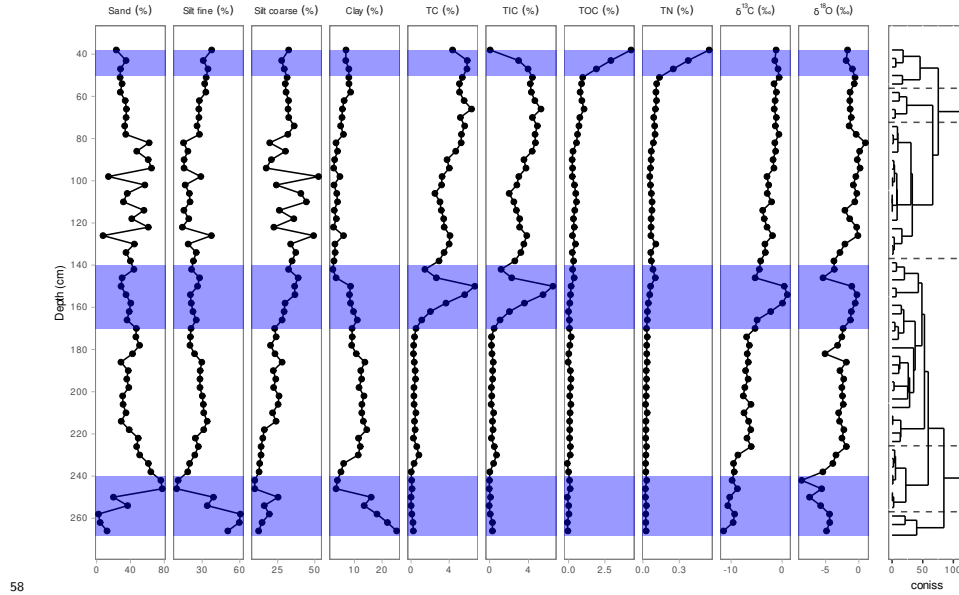
28 Elemental geochemical analyses were performed every 1 cm for total carbon
 29 (TC), total inorganic carbon (TIC), total organic carbon ($TOC = T - TIC$) and
 30 total sulfur (TS) in a LECO SC 144 DR furnace (IPE-CSIC). Total nitrogen
 31 (TN) was measured in a VARIO MAX CN elemental analyzer (IPE-CSIC).
 32 Grain-size distributions between 0.02 μm and 2000 μm were measured using
 33 a Malvern Mastersizer 2000 APA2000 grain-size analyzer with Hydro 2000G
 34 AWA2000. Sedimentary facies were described based on criteria formulated by
 35 (Schnurrenberger et al., 2003) and lacustrine units have been classified according
 36 to the sedimentary structure, MSCL data, elemental composition (TOC, TIC,
 37 TN, TS, TOC/TN, BioSi), XRF data and waxes composition.

38 *3.2. Chronology*

39 The chronology for the Miscanti (MIS13-2A) and Miñique (MIN13-1F) se-
 40 quence was obtained using ^{210}Pb and ^{137}Cs dating techniques, eight U/Th dates
 41 and, nine ^{14}C AMS dates (see Table 1). Core MIN13-1F-1G was sampled in
 42 the field every 0.5 and 1 cm for ^{210}Pb and ^{137}Cs . The MIS13-2A-1G sequence
 43 was correlated with the ^{210}Pb and ^{137}Cs data from the sequence obtained by
 44 Valero-Garcés et al., 1996. The radiocarbon ages were determined in MIS13-
 45 2A-1G and MIN13-1F-1G cores by AMS at Poznan Radiocarbon Laboratory,
 46 Poland and the DirectAMS laboratories, USA. Waters from the hypolimnion has
 47 also sampled to assess modern ^{14}C reservoir effects (Table 1). The U/Th dating
 48 was done at the Massachusetts Institute of Technology (MIT, McGee Lab). For
 49 U/Th samples, the cores were visually assessed for material to dating criteria and
 50 methods described in (Chen et al., 2020; Lawrence Edwards et al., 1987; Shen et
 51 al., 2002). A VG PQ2+ quadrupole ICP-MS and an Agilent 7900 ICP-MS at
 52 MIT was using to determined uranium and thorium concentration. Samples with
 53 higher U/Th ratios were then identified as materials worth further processing

as they are more likely to yield “clean” samples with high $^{230}\text{Th}/^{232}\text{Th}$ ratios. The isotopic compositions of the resulting uranium and thorium fractions were then measured on a Nu Plasma II-ES multicollector ICP-MS at MIT.

3.3. Data analysis



4. Results

4.1. Cores correlations and sedimentary proxies

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